

End-to-end Automatic Body and Face Setup for Generative or User Created 3D Avatar

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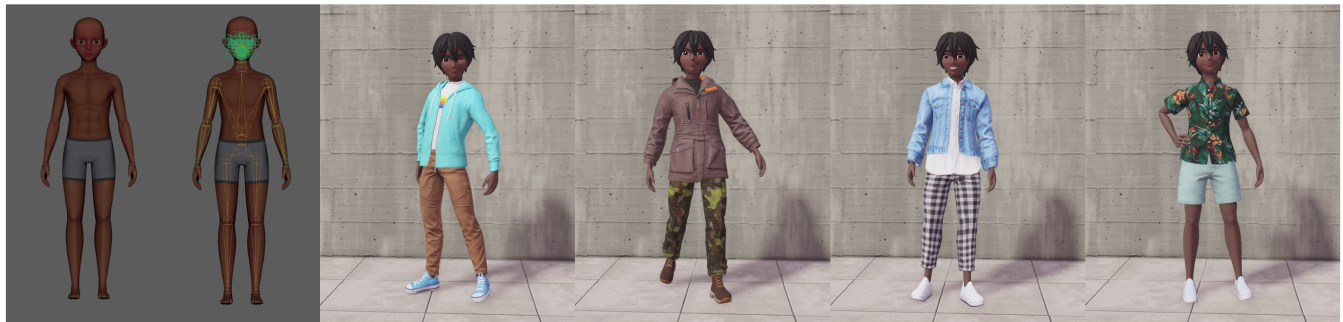


Figure 1: We introduce a pipeline to automatically rig and cage an input model to make it ready for animation (including facial animation) and different customization (clothing, hair and shoes).

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1 INTRODUCTION

With recent advances in generative AI and other 3D content creation technologies, the process of creating 3D models is becoming less difficult. However, creating animatable and customizable avatars is still a significant effort, especially when facial animation and customizable clothing/accessories are required. In this talk,

we introduce a production pipeline that enables users with limited expertise to create and customize avatars. Our fully automatic system is able to convert an input geometry-only model into an animation-ready, rigged and skinned avatar with facial animations and customizable with clothing and accessories. We implemented a multi-stage pipeline that combines machine learning and geometry processing techniques. The pipeline can:

- handle a large variety of inputs without restriction to high quality meshes (water-tight, manifold etc),
- detect and segment the head from the body,
- predict facial expressions and generate a facial rig,
- and infer a low-resolution, standardized cage around the geometry which enables users to customize clothing and accessories.

2 PIPELINE OVERVIEW

Our pipeline gives creators an opportunity to produce gameplay and facial chat-ready avatars from generic 3D humanoid-like models, which can be obtained from generative ML or digital marketplaces. The output is a fully rigged, skinned, caged and facially animated avatar that can be used across a variety of 3D experiences.

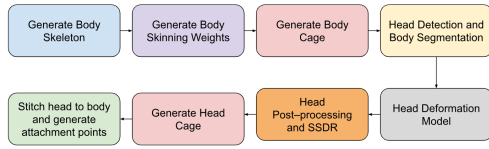


Figure 2: A simplified diagram of our pipeline

2.1 Automatic Body Rigging and Skinning

We first decimate and remesh the models to handle input with different qualities. Inspired by the work of RigNet [Xu et al. 2020], we use graph neural networks and edge convolutions to regress each vertex and form potential groups for the joints. To construct a proper skeleton, we perform k-means clustering to find the center of each group to compute joints. Then, we heuristically connect all the joints into a template skeleton structure. Compared with the original RigNet, our solution does not enforce symmetry in the skeleton generation, which ends up providing better results for a larger variety of inputs from production.

For skinning, we use a similar network architecture to regress skinning weights on each vertex based on the skeleton input we predicted previously. We use the vertex positions and geodesic distance to all joints as features to train our skinning model. We conduct postprocessing to clean up and smooth the weights to make it animation ready.

2.2 Automatic Face Rigging and Skinning

Face rigging takes a head as input with neutral expression and outputs a FACS (facial action coding system) based LBS rig. Facial rigging is performed using a three step approach. First, an ML deformation model predicts 105 FACS poses of the neutral mesh. Second, we apply postprocessing to resolve collisions in predicted poses. Finally, we apply SSDR [Le and Deng 2012] to convert the pose shapes into an LBS rig. The deformation model and post processing steps are agnostic to the rig. Only the SSDR step is rig specific. This means that we can swap out the final step to create a blendshape rig or change the number of joints in the SSDR optimization to trade off fidelity for speed.

The core piece of this pipeline is an ML model that predicts the deformations for each pose. The model takes a neutral mesh and a one-hot encoded FACS pose vector as input and predicts 3d vertex displacements to deform the neutral mesh into the pose. The architecture is based on spatial diffusion layers introduced by DiffusionNet [Sharp et al. 2022]. We add an additional conditional input (FACS-vector) and introduce a global feature encoder sub-network that allows for global shape information to flow into the deformation prediction.

2.3 Auto-Caging

Our avatars use a caging system to allow users to create 3D clothing and accessories that any avatar can wear with any combination of other clothing in layers. A body cage is an envelope that closely wraps around the body mesh with a standardized fixed topology.

To automatically generate the body cages, we use a PointNet-based [Qi et al. 2017] solution to regress the body cage vertex



Figure 3: FACS poses generated by deformation model

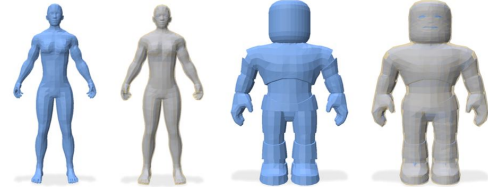


Figure 4: Body (left) and cage overlay (right) in each pair

positions given sampled point clouds on the body mesh as input. The output of the ML model provides cage vertex positions. We perform ARAP [Sorkine and Alexa 2007] as postprocessing to pull out the predicted cage vertices that are under the body mesh.

3 DATA, IMPLEMENTATION AND EVALUATION

Our training data is a mix of avatars created for our platform and synthetic augmented data. To obtain high quality 3D models for our training data, we developed many new technologies to improve the data acquisition and augmentation process. Tools include interpolation and extrapolation with scripted deformations on salient areas of the existing 3D models. We use roughly 10K models for body training and 4K models for the face. Our pipeline has been successfully integrated into an authoring tool as a backend service, and the runtime of the entire pipeline is approximately 107 seconds. We evaluate the results with various metrics and feedback from experienced 3D artists.

4 CONCLUSION

We implemented a production level end-to-end pipeline to automatically bridge the gap between 3D humanoid models and body and facial animation-ready avatars that are highly customizable with clothing and accessories. Our current focus has been handling bipeds and humanoid models, but in the future, we plan to expand the range to include quadrupeds and non-humanoid input.

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